

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456470.87 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.043
Transit depth (Rp/Rs)^2	1.07 +/- 0.88 [%]
Orbital Inclination (inc)	85.01 +/- 2.26
Airmass coefficient 1 (a1)	0.689 +/- 0.012
Airmass coefficient 2 (a2)	0.325 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#2 - [479, 50]
Optimal Aperture	3.34
Optimal Annulus	13.72
Transit Duration (day)	0.063 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.103 ± 0.043 vs 0.1326 (deviation 0.69σ)
Duration fit vs published	0.063 d vs 0.0737 d (deviation 0.37σ)
Mid-time O–C	-7.4 min
Depth SNR	2.4
Residual scatter	1.76 %

Light curve

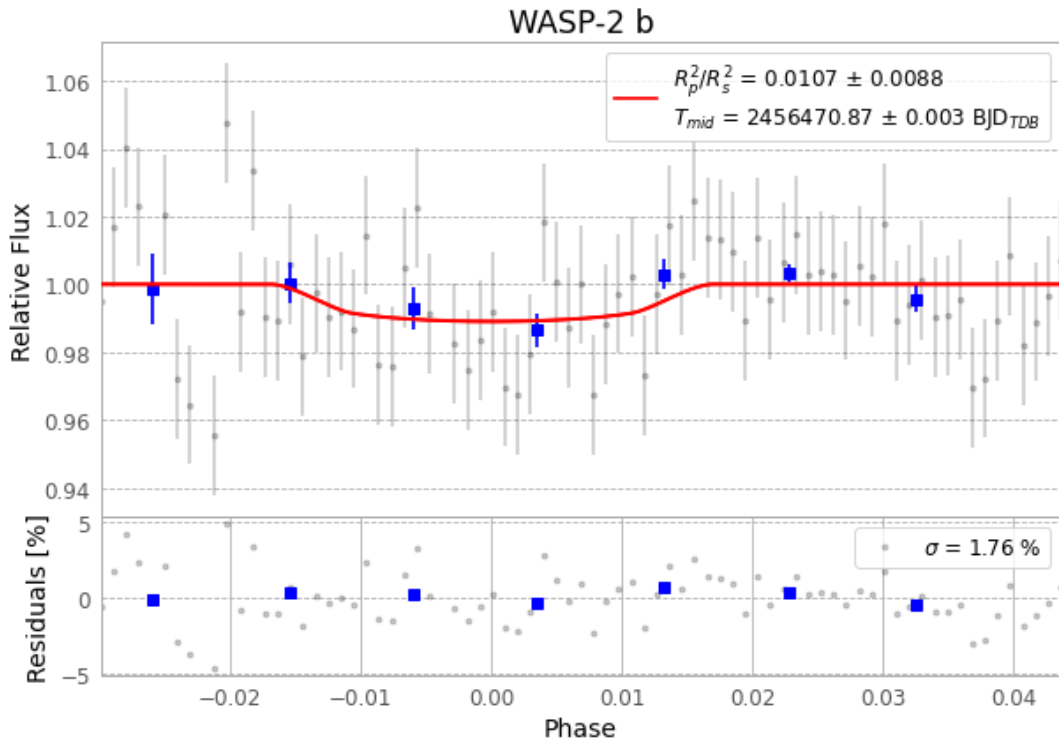


Figure 1 — FinalLightCurve_WASP-2 b_2013-06-27.png (SHA-256 0280e490c385607c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8633a025c2253d43...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-2130627071212.FITS → WASP-2130627110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130627.log (c54734aca8de...), FinalLightCurve_WASP-2 b_2013-06-27.png (0280e490c385...), FinalLightCurve_WASP-2 b_2013-06-27.csv (e95935b61008...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:23Z.